

Darren Zheng

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EDUCATION

University of Pennsylvania

Aug 2021 — May 2025

Bachelor of Arts in Mathematics, Minor in Statistics & Data Science, Ben Franklin Scholar *B.A. GPA: 3.51/4.0*

- Relevant Coursework: Data Analytics and Statistical Computing, Intro to Optimization, Corporate Finance, Abstract Probability Theory, Measure Theory, Stochastic Processes, Partial Differential Equations

PROJECTS

Undergraduate Honor's Thesis | [Link to PDF](#)

June 2024 — May 2025

- Constructed intrinsic Morse functions on Grassmannian manifolds to compute its Morse homology under Prof. Dennis DeTurck

Directed Reading Program | [Link to Presentations](#)

Aug 2022 — May 2025

- Semester-long independent study with a graduate student. Prepared a presentation/write-up at the end of each semester to rest of participants and mentors
- Topics: Classification of Hyperbolic Knots, Generating Functions, Pontryagin Duality, Brownian Motion, Erdős Set Distance Problem, Falconer's Conjecture, Mapping Class Groups

EWS Detection using Topological Data Analysis | [Link to PDF](#)

Jan 2024 — Apr 2024

- Applied a proposed persistent-homology-based methodology from a paper by Gidea & Katz in Python & R to study the time-series of certain stock market indicators as potential early warning signs of market collapse

ACTIVITIES

Penn Math Contest Club | *Co-founder & President*

Aug 2021 — May 2025

- Co-founded new club during freshmen year, focused on hosting a math contest for high schoolers
- Helped write and quality-check ~120 original problems per year for the test, organized day-of logistics, operated social outreach with local schools, and secured funding from sponsors
- Grew club from 40 participants (online due to COVID) to ~150 in-person and ~220 online participants in Spring 2025; [More details on website here](#)

Penn Undergraduate Math Society | *President*

Jan 2021 — May 2025

- Joined board as freshmen liaison, president during senior year. Ran events for all math-adjacent undergraduates
- Events include: Course & Major Advising, Graduate Student Talks, Integration Bee, Department Merch Sale, Alumni Career Insights, Weekly Office Hours, DRP × PUMS Mini-Talks, Faculty Lunches, Industry Recruiting

Science Olympiad at Penn | *Test Writer & Event Supervisor*

Aug 2021 — May 2025

- Wrote 7 tests and acted as event supervisor across It's About Time, Fermi Problems, Optics, and Codebusters categories for high school competitors over 4 years
- Joined finance board during 2023 — 2024 to secure sponsorship funding for the high-school contest

Student Activities Council | *Executive Board*

Feb 2023 — Dec 2024

- Managed the approval and allocation of a \$1.5 million budget for over 300 student groups
- Led part of the New Group Recognition Process each fall and spearheaded improvements to club debt policies

Putnam Club | *Organizer*

Aug 2021 — Dec 2024

- Organized club that meets weekly on Sundays in the fall to practice for the Putnam Exam

TEACHING

University of Pennsylvania Math Department | *Teaching Assistant*

Aug 2022 — May 2025

- Was responsible for grading assignments/exams, weekly office hours, administering weekly quizzes and recitations
- Classes included: Calculus II (x2), Honors Calculus I, Honors Calculus II, Abstract Linear Algebra

Ross Mathematics Program | *Administrator*

June 2024 — July 2024

- Was an administrator during the 6-week math camp for ~80 high schoolers, ~12 counselors, ~5 professors
- Handled various organizational tasks, coordinating social events for the students, and preparing various mathematical talks, including an week-long introduction to point set topology talk series

AwesomeMath Summer Program | *Teaching Assistant*

June 2022 / June 2023 / June 2025

- Was a TA for multiple summers. Did 2 courses per each 3-week session for 9 weeks each summer
- Hosted problem sessions and graded daily HW for classes between 10-30 students
- Taught primarily combinatorics I/II, algebra III/IV, and number theory III/IV

SKILLS & TECHNOLOGIES

Languages: Python (Numpy, Pandas), R, Matlab, Java, JavaScript, Go, LaTeX, Excel